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Laplace Operator in Polar Coordinates

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PARTIAL DIFFERENTIAL EQUATIONS

AN INTRODUCTION

WALTER A. STRAUSS
Brown University

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We want to show that $u \equiv v$ in D. So we simply subtract equations and let $w = u - v$. Then $\Delta w = 0$ in D and $w = 0$ on bdy D. By the maximum principle

$$0 = w(\mathbf{x}_m) \le w(\mathbf{x}) \le w(\mathbf{x}_M) = 0 \quad \text{for all } \mathbf{x} \in D.$$

Therefore, both the maximum and minimum of $w(\mathbf{x})$ are zero. This means that $w \equiv 0$ and $u \equiv v$.

INVARIANCE IN TWO DIMENSIONS

The Laplace equation is invariant under all rigid motions. A rigid motion in the plane consists of translations and rotations. A *translation* in the plane is a transformation

$$x' = x + a \qquad y' = y + b.$$

Invariance under translations means simply that $u_{xx} + u_{yy} = u_{x'x'} + u_{y'y'}$.

A *rotation* in the plane through the angle α is given by

$$\begin{aligned} x' &= x\cos\alpha + y\sin\alpha \\ y' &= -x\sin\alpha + y\cos\alpha. \end{aligned} \tag{4}$$

By the chain rule we calculate

$$\begin{aligned} u_x &= u_{x'}\cos\alpha - u_{y'}\sin\alpha \\ u_y &= u_{x'}\sin\alpha + u_{y'}\cos\alpha \\ u_{xx} &= (u_{x'}\cos\alpha - u_{y'}\sin\alpha)_{x'}\cos\alpha - (u_{x'}\cos\alpha - u_{y'}\sin\alpha)_{y'}\sin\alpha \\ u_{yy} &= (u_{x'}\sin\alpha + u_{y'}\cos\alpha)_{x'}\sin\alpha + (u_{x'}\sin\alpha + u_{y'}\cos\alpha)_{y'}\cos\alpha. \end{aligned}$$

Adding, we have

$$\begin{aligned} u_{xx} + u_{yy} &= (u_{x'x'} + u_{y'y'})(\cos^2\alpha + \sin^2\alpha) + u_{x'y'}\cdot(0) \\ &= u_{x'x'} + u_{y'y'}. \end{aligned}$$

This proves the invariance of the Laplace operator. In engineering the laplacian Δ is a model for *isotropic* physical situations, in which there is no preferred direction.

The rotational invariance suggests that the two-dimensional laplacian

$$\Delta_2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$$

start

should take a particularly simple form in *polar coordinates*. The transformation

$$x = r\cos\theta \qquad y = r\sin\theta$$

has the jacobian matrix

$$\mathcal{J} = \begin{pmatrix} \frac{\partial x}{\partial r} & \frac{\partial y}{\partial r} \\ \frac{\partial x}{\partial \theta} & \frac{\partial y}{\partial \theta} \end{pmatrix} =$$

with the inverse matrix

$$\mathcal{J}^{-1} = \begin{pmatrix} \frac{\partial r}{\partial x} & \frac{\partial \theta}{\partial x} \\ \frac{\partial r}{\partial y} & \frac{\partial \theta}{\partial y} \end{pmatrix}$$

(Beware, however, that $\partial r/\partial x \neq ($

$$\frac{\partial}{\partial x} = \cos$$

$$\frac{\partial}{\partial y} = \sin$$

These *operators* are squared to gi

$$\frac{\partial^2}{\partial x^2} = \left[\cos\theta\frac{\partial}{\partial r} - \right.$$

$$= \cos^2\theta\frac{\partial^2}{\partial r^2} -$$

$$+ \frac{\sin^2\theta}{r^2}\frac{\partial^2}{\partial\theta^2}$$

$$\frac{\partial^2}{\partial y^2} = \left(\sin\theta\frac{\partial}{\partial r} + \right.$$

$$= \sin^2\theta\frac{\partial^2}{\partial r^2} +$$

$$+ \frac{\cos^2\theta}{r^2}\frac{\partial^2}{\partial\theta}$$

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angle α is given by

$+ y \sin\alpha$ (4)
$+ y \cos\alpha.$

$\alpha - (u_{x'} \cos\alpha - u_{y'} \sin\alpha)_{y'} \sin\alpha$
$\alpha + (u_{x'} \sin\alpha + u_{y'} \cos\alpha)_{y'} \cos\alpha.$

$\cos^2\alpha + \sin^2\alpha) + u_{x'y'} \cdot (0)$

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with the inverse matrix

$$\mathcal{J}^{-1} = \begin{pmatrix} \dfrac{\partial r}{\partial x} & \dfrac{\partial \theta}{\partial x} \\ \dfrac{\partial r}{\partial y} & \dfrac{\partial \theta}{\partial y} \end{pmatrix} = \begin{pmatrix} \cos\theta & \dfrac{-\sin\theta}{r} \\ \sin\theta & \dfrac{\cos\theta}{r} \end{pmatrix}.$$

(Beware, however, that $\partial r/\partial x \neq (\partial x/\partial r)^{-1}$.) So by the chain rule we have

$$\frac{\partial}{\partial x} = \cos\theta \frac{\partial}{\partial r} - \frac{\sin\theta}{r}\frac{\partial}{\partial \theta},$$
$$\frac{\partial}{\partial y} = \sin\theta \frac{\partial}{\partial r} + \frac{\cos\theta}{r}\frac{\partial}{\partial \theta}.$$

These *operators* are squared to give

$$\begin{aligned}
\frac{\partial^2}{\partial x^2} &= \left[\cos\theta \frac{\partial}{\partial r} - \frac{\sin\theta}{r}\frac{\partial}{\partial \theta}\right]^2 \\
&= \cos^2\theta \frac{\partial^2}{\partial r^2} - 2\left(\frac{\sin\theta\cos\theta}{r}\right)\frac{\partial^2}{\partial r \partial \theta} \\
&\quad + \frac{\sin^2\theta}{r^2}\frac{\partial^2}{\partial \theta^2} + \frac{2\sin\theta\cos\theta}{r^2}\frac{\partial}{\partial \theta} + \frac{\sin^2\theta}{r}\frac{\partial}{\partial r} \\
\frac{\partial^2}{\partial y^2} &= \left(\sin\theta \frac{\partial}{\partial r} + \frac{\cos\theta}{r}\frac{\partial}{\partial \theta}\right)^2 \\
&= \sin^2\theta \frac{\partial^2}{\partial r^2} + 2\left(\frac{\sin\theta\cos\theta}{r}\right)\frac{\partial^2}{\partial r \partial \theta} \\
&\quad + \frac{\cos^2\theta}{r^2}\frac{\partial^2}{\partial \theta^2} - \frac{2\sin\theta\cos\theta}{r^2}\frac{\partial}{\partial \theta} + \frac{\cos^2\theta}{r}\frac{\partial}{\partial r}.
\end{aligned}$$

(The last two terms come from differentiation of the coefficients.) Adding these operators, we get (lo and behold!)

$$\boxed{\Delta_2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} = \frac{\partial^2}{\partial r^2} + \frac{1}{r}\frac{\partial}{\partial r} + \frac{1}{r^2}\frac{\partial^2}{\partial \theta^2}.} \tag{5}$$

It is also natural to look for special *harmonic functions that themselves are rotationally invariant*. In two dimensions this means that we use polar